FTP AND SNMP NETWORK MANAGEMENT

Practical Assignment — Questions and Answers

1. Install FileZilla Server, Create a User and Share a Folder

Question 1: Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.

Answer

File Transfer Protocol (FTP) is used to transfer files between a client and a server over a network. FileZilla Server provides an FTP service, while FileZilla Client is used to connect to it and transfer files.

Installation and initial setup

1. Download FileZilla Server from the official FileZilla website and run the installer. Choose the standard installation options suitable for your Windows computer.
2. Complete the setup and open the FileZilla Server administration interface. If prompted, connect to the local server administration service.
3. Create a folder for shared files, for example C:\FTPShare. Make sure the Windows account running FileZilla Server has permission to access this folder.
4. In the FileZilla Server administration interface, open the Users section and add a new user, for example musicuser. Enable password authentication and set a strong password.

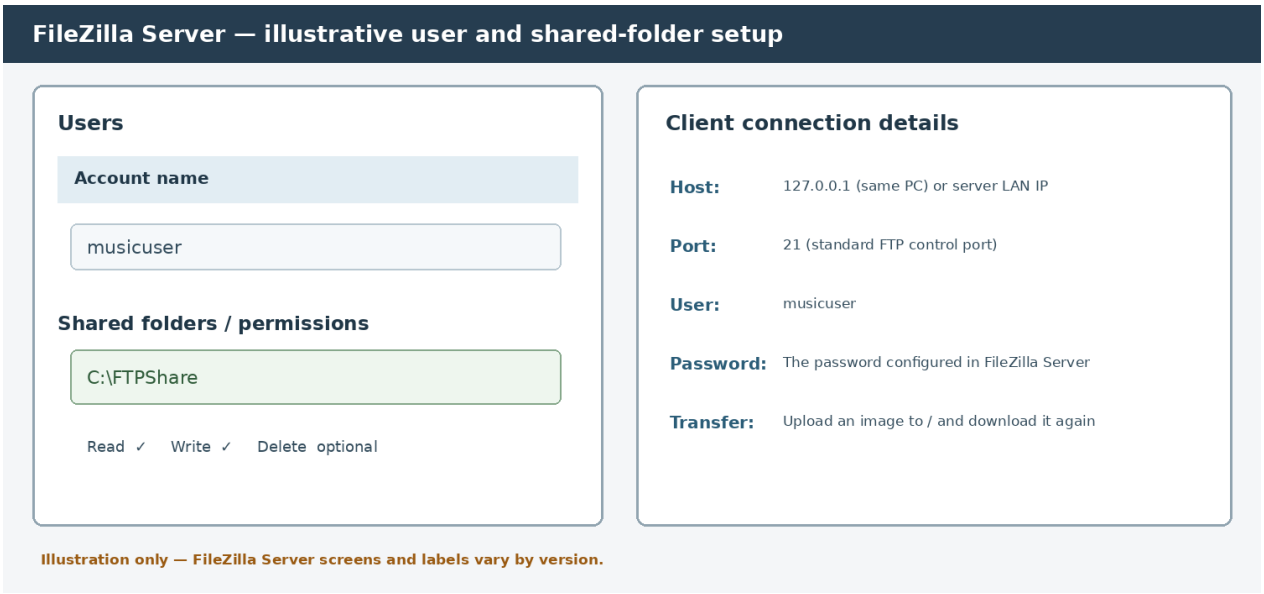


Figure 1. Illustrative FileZilla Server user and shared-folder setup. The actual interface may differ by version.

1. FileZilla Server — Shared Folder and Permissions

Configure the shared folder

5. Open the user's Shared folders or Mount points settings and add the physical folder C:\FTPShare.
6. Assign the user a virtual path, commonly /, so the shared folder appears as the root directory after login.
7. Grant Read permission so the user can list and download files. Grant Write permission so the user can upload files. Enable Delete or Append only if your task requires them.
8. Save or apply the settings. Confirm that the server is listening for FTP connections. The standard FTP control port is TCP 21 unless you deliberately configure a different port.

Network and firewall checks

- For a same-computer test, use 127.0.0.1 as the host.
- For another computer on the same LAN, use the Windows machine's LAN IPv4 address and allow the FTP server through Windows Defender Firewall.
- For a basic LAN lab, use plain FTP only on a trusted isolated network. Standard FTP does not encrypt usernames, passwords or file contents. For real-world use, prefer FTPS or SFTP where supported.

Conclusion

The server is ready when the user account can authenticate and can access the configured shared directory with the assigned permissions.

2. Connect with FileZilla Client and Upload an Image

Question 2: Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.

Answer

FileZilla Client provides a graphical interface for connecting to FTP servers. The left side normally shows files on the local computer, while the right side shows files on the remote FTP server.

Connection procedure

9. Install and open FileZilla Client on the same computer or another computer on the same network.
10. Enter the server Host (127.0.0.1 for a same-PC test, or the FTP server's LAN IP), Username (musicuser), Password (the password created on the server) and Port 21.
11. Click Quickconnect. If a connection is successful, the right-hand Remote site panel displays the shared folder.
12. In the left-hand Local site panel, browse to an image file such as a JPG or PNG.
13. Select the image and double-click it, or right-click and choose Upload. Wait for the transfer queue to show that the transfer completed successfully.

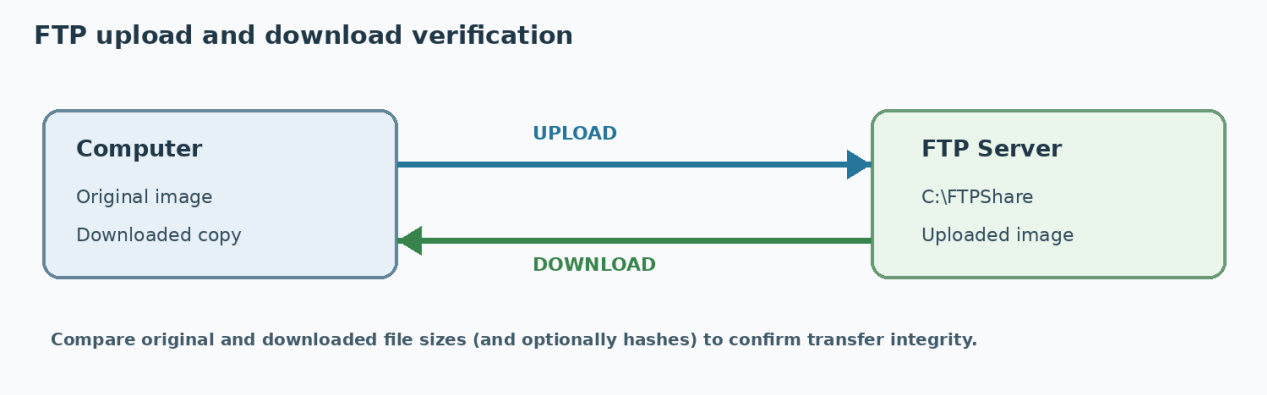


Figure 2. Upload and download flow between a local computer and an FTP server.

Expected observation

The uploaded image should appear in the Remote site listing. The transfer queue should show a successful transfer rather than a failed or skipped transfer.

3. Download the File and Verify Its Size

Question 3: Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.

Answer

Download procedure

14. Create a separate destination folder on your computer, for example C:\FTPDownload. Do not use the original image folder.
15. In FileZilla Client, browse to C:\FTPDownload in the Local site panel.
16. On the Remote site panel, locate the image uploaded in Task 2.
17. Right-click the remote image and choose Download, or double-click it. Wait for the transfer queue to report success.
18. Open File Explorer and compare the original image file with the downloaded copy.

Verify file size

In Windows File Explorer, right-click each file, select Properties and compare the Size value. The downloaded file should have the same size as the original. You can also compare hashes using PowerShell:

```
Get-FileHash "C:\Path\To\Original\image.jpg" -Algorithm SHA256
Get-FileHash "C:\FTPDownload\image.jpg" -Algorithm SHA256
```

If the SHA256 hash values are identical, the files have identical contents. File sizes matching is a useful basic check, while matching hashes provide stronger confirmation.

Result table

Check	Original image	Downloaded copy
File name	image.jpg	image.jpg
File size	_____ bytes	_____ bytes
SHA256 hash (optional)	_____	_____

Conclusion

A successful download and matching file size show that the transfer completed as expected. Matching SHA256 hashes confirm that the downloaded file is byte-for-byte identical to the original.

4. Three Real-World Scenarios Where SNMP Is Used

Question 4: Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use, such as ISPs, campus Wi-Fi, or large websites. Explain how network administrators monitor device health, bandwidth or uptime.

Answer

Simple Network Management Protocol (SNMP) allows a network management system to collect information from SNMP-enabled routers, switches, access points, firewalls and other devices. Administrators use this information to monitor performance, detect faults and plan network capacity.

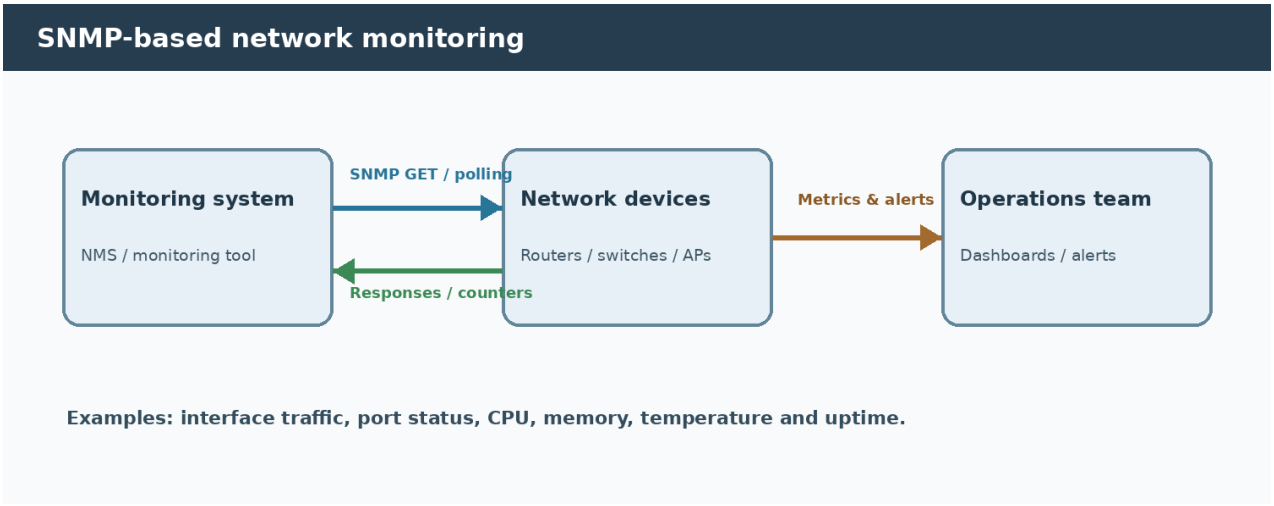


Figure 3. Simplified SNMP monitoring model: a management system polls network devices and presents metrics to administrators.

Scenario 1 — Internet Service Provider (ISP)

An ISP operates routers, switches and customer-edge equipment across its network. A network operations team can use SNMP polling to monitor interface traffic counters, bandwidth utilization, port status, errors and device uptime. If a link becomes congested or an interface goes down, the monitoring platform can alert engineers so they can investigate before customers experience prolonged slow service or an outage.

Scenario 2 — Campus or public Wi-Fi

A university, college or public venue may operate many wireless access points and network switches. SNMP can help administrators monitor whether access points and switches are online, how much traffic each interface carries, and whether devices report hardware or resource problems. This helps identify overloaded uplinks, failed access points or areas where capacity may need to be increased.

Scenario 3 — Large website or cloud/data-centre infrastructure

A large online service depends on data-centre switches, routers and firewalls to connect servers and storage systems. SNMP monitoring can track device availability, CPU and memory health where exposed, interface throughput, errors and temperature or power status on supported hardware. Operations teams use dashboards and alerts to identify network bottlenecks or failing equipment that could affect website availability.

4. SNMP Scenarios — Benefits and Research Notes

How SNMP supports administrators

- Polling: The monitoring system periodically requests management values from network devices.
- MIB and OID: A Management Information Base describes the managed values, and Object Identifiers identify individual metrics.
- Traps/informs: Devices can send notifications when important events occur, such as link changes or threshold events.
- Trend analysis: Collected counters help teams identify recurring congestion and plan upgrades.

SNMP is a monitoring protocol; it does not itself guarantee that an application or website is available. Network teams often combine SNMP with application checks, logs and end-to-end tests.

Security note

SNMPv1 and SNMPv2c use community strings and do not provide strong encryption. SNMPv3 supports authentication and privacy features and is generally preferred for secure management. Administrators should restrict which monitoring systems can query devices and avoid exposing management interfaces to untrusted networks.

Research sources

The following technical documentation supports the use cases and metrics described above:

• Grafana Labs — SNMP integration: device interface traffic, errors, status, CPU, memory, uptime and hardware status.

• Cisco Meraki — SNMP overview and configuration: polling access points, switches and security appliances for device status and interface packet/byte counters.

• Paessler PRTG — Monitoring via SNMP: bandwidth usage, uptime, traffic levels, CPU and memory monitoring.

• Datadog — Network Device Monitoring: device discovery, bandwidth utilization, bytes sent and device up/down status.

These are examples of capabilities available in network-management deployments; they do not claim that a specific consumer app personally uses SNMP in its own production network.

5. Comparison Between FTP and HTTP

Question 5: Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.

Answer

FTP (File Transfer Protocol) is designed specifically for transferring files between a client and server. HTTP (Hypertext Transfer Protocol) is designed for transferring web resources such as HTML pages, images, scripts and API data.

Comparison point	FTP	HTTP
Default port number	TCP 21 for control; traditional active/passive data connections use additional ports.	TCP 80 by default. HTTPS commonly uses TCP 443; HTTP/3 uses QUIC over UDP 443.
Security	Plain FTP does not encrypt credentials or file data. FTPS adds TLS; SFTP is a different protocol that runs over SSH.	Plain HTTP is not encrypted. HTTPS protects HTTP using TLS.
Data transfer method	Uses FTP commands and separate control/data connections; active and passive modes are supported.	Uses request-response messages. Client sends a request and server returns a response containing a resource or status.
Main use cases	Uploading, downloading and managing files on a remote file server; website file publishing and batch file exchange.	Browsing websites, loading web resources, communicating with REST APIs and transferring application data.
Authentication	Can use username/password; anonymous access is also possible if configured. Plain FTP credentials are exposed on the network.	Can be anonymous/public, or use cookies, tokens, HTTP authentication or application login mechanisms. HTTPS protects credentials in transit.
Typical client	FileZilla Client, command-line FTP clients and automated transfer tools.	Web browsers, mobile apps, API clients and command-line tools such as curl.
Connection behavior	Dedicated file-transfer protocol with directory listing and file operation commands.	General-purpose web protocol supporting methods such as GET, POST, PUT and DELETE.

Conclusion

FTP is specialized for file transfers and remote file management, while HTTP is the foundation of web browsing and web APIs. Both have secure variants or secure deployment options: FTPS for FTP and HTTPS for HTTP. For sensitive transfers, avoid plain FTP and use encrypted protocols.

Final Conclusion and Practical Screenshot Checklist

This assignment demonstrates the use of FTP for transferring files and SNMP for monitoring network infrastructure. FileZilla Server provides a shared FTP location, FileZilla Client uploads and downloads files, and file size or hash comparisons help verify transfer accuracy. SNMP allows network administrators to monitor devices, traffic and uptime. The final comparison highlights the different purposes and security characteristics of FTP and HTTP.

Screenshots to include in your submission

- Task 1: FileZilla Server showing the created user and shared-folder permissions.
- Task 2: FileZilla Client showing a successful connection and uploaded image in the Remote site panel.
- Task 3: FileZilla Client showing the downloaded file, plus Windows Properties or PowerShell output comparing file size or SHA256.
- Task 4: The SNMP monitoring diagram may be used as an explanatory visual; cite the technical sources listed in the assignment.
- Task 5: The FTP vs HTTP comparison table is included above.

The FileZilla and SNMP diagrams in this document are illustrative teaching aids, not screenshots from an actual installation or monitoring system. Replace them with screenshots from your own Windows computer where the instructor requires practical evidence.